



Week 8 Lecture 2

▼ Class	BSCCS2003
🕒 Created	@October 24, 2021 10:58 AM
🔗 Materials	
# Module #	45
▼ Type	Lecture
# Week #	8

Asynchronous Updates

Original Web

- Client send request → Server responds → Client displays
- For any update of the page
 - A new request sent from the client to the server
 - Server has to respond with complete page, HTML, styling, etc
 - Client has to render the page again from scratch
- Potential issues
 - Server load → lots of redundant data to be sent each time
 - Server-rendering → More work
 - Slow updates → Load full page, re-render

Asynchronous Updates

- Update only part of the page
 - Load extra data in the background after the main page has been loaded and rendered
- Quick response on the main page → Better user experience
- Request for update can ask for just minimal data to refresh part of the page
 - Example → Show user a form to select animal
 - Request data about animal alone from the server → No need for HTML or other styling

- Refresh only one `<div>` in the page with text about the animal
- Originally seen as AJAX, now many variants

Core idea → Refresh part of the document based on asynchronous (background) queries to the server

DOM

Document Object Model

- Programming interface for web documents
- What is a web-page?
 - HTML source? Rendered image?
- DOM is an abstract model (tree structure) of the document
- Object-oriented allows manipulation like known objects
- Tightly coupled with JavaScript in most cases
 - Can also be manipulated from other languages (Python has XML DOM interface for example)

Example Usage

https://developer.mozilla.org/en-US/docs/Web/API/Document_Object_Model/Introduction

```
const paragraphs = document.querySelectorAll("p");
// paragraphs[0] is the first <p> element
// paragraphs[1] is the second <p> element, etc.
alert(paragraphs[0].nodeName);
```

Manipulating the DOM

```
<html>
  <head>
    <script>
      // Run this function when the document is loaded
      window.onload = function() {
        // Create a couple of elements in an otherwise empty HTML page
        const heading = document.createElement("h1");
        const heading_text = document.createTextNode("Big Head!");
        heading.appendChild(heading_text);
        document.body.appendChild(heading);
      }
    </script>
  </head>
  <body>
  </body>
</html>
```

Component building

- DOM is manipulatable through programs
- Bring concepts of programming into DOM manipulation:
 - Objects
 - Composition of objects, inheritance
 - Loops, iterators, programmable placement
- Lot more flexibility in front-end
- Lot more complexity in the front-end

Summary

- Asynchronous updates opened up front-end development
- Many new frameworks, technologies